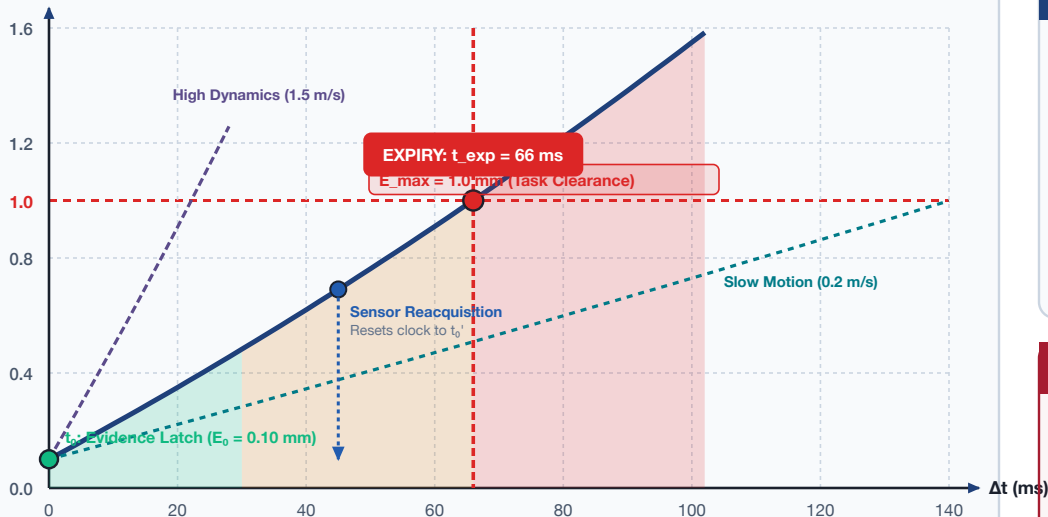


TEMPORAL BELIEF UNCERTAINTY GROWTH & INVALIDATION HORIZON

Bounding Covariance Expansion $E(t) = E(0) + 3\sigma_v t + \frac{1}{2} a_{\text{dist}} t^2$ Against Mechanical Task Clearance E_{max}

Error Bound $E(t)$ [mm]



VALID [0–30 ms]

Full Authority

DEGRADED [30–66 ms]

Slowdown / Widen Buffer

EXPIRED [>66 ms]

Action Perms Revoked

GOVERNING DECAY LAW

$$E(t) = E(0) + 3\sigma_v t + \frac{1}{2} a_{\text{dist}} t^2$$

$$t_{\text{exp}} = \inf \{ t \geq 0 : E(t) \geq E_{\text{max}} \}$$

- **Initial Sensor Bound $E(0)$:**
0.10 mm at P99.9 metrology latch
- **Linear Velocity Drift $3\sigma_v$:**
12.0 mm/s (unobserved creep)
- **Disturbance Accel a_{dist} :**
50.0 mm/s² (friction/backlash)
- **Empirical Characterization:**
Fitted from staged P99 dropouts

THE FROZEN TRANSFORM TRAP

- × **The Software Illusion:**
Middleware republishes cached pose with fresh OS clock ($t_{\text{pub}} = t_{\text{now}}$).
- × **Deceptive Freshness:**
Downstream age check ($t_{\text{now}} - t_{\text{pub}} < 2$ ms) passes with zero warnings.
- × **Physical Reality:**
True evidence age $\Delta t > 66$ ms; physical error $E(t) \gg E_{\text{max}}$ (crash hazard).
- × **Architectural Invariant:**
Age MUST bind strictly to acquisition epoch t_0 , never to publish time.